

Manthan Gadhia

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Zürich, Switzerland

MSc Science, Technology, and Policy

MSc student with an interdisciplinary training in the theoretical fundamentals of AI and practical considerations of human-centred ML and data science applications with a special interest in NLP. Interested in societal impacts of AI and computational technologies on labour, education, politics, and social networks.

Education

ETH ZÜRICH, CH

MSc, Science, Technology, and
Policy (CS focus)
2026 (expected)

RADBOD UNIVERSITY, NL

BSc, Artificial Intelligence
2023

Skills and competencies

Programming: Python, R

ML Infrastructure: PyTorch,
HuggingFace, training
pipelines

NLP: Statistical analysis
methods, RNNs/LSTMs,
Transformers, RAG

**High-performance
computing:** SLURM, Docker

Relevant Coursework

BSc: NLP; Deep Learning;
Societal impacts of AI; Data in
Culture and Society; HCI

MSc: Intro/Advanced ML;
Interactive ML; LLMs for Law;
Design in EdTech; AI for
Education; AI for Human
Values; AI personhood and
social justice; Social Data
Science; Intro to Social
Network Analysis

Experience

MSC THESIS GENDER BIAS IN LLM POST-TRAINING

Jan 2026 – Jul 2026 (expected)

Analysing development of gender biases in open-source LLMs through their post-training stages when models infer the gender of individuals given demographic specifications. I observe an over-correction towards female-preference after going through alignment.

STUDENT ASSISTANT WELLBEING, INCLUSION, DIVERSITY, AND EQUITY

Nov 2023 – Present

Organising topic-relevant events and communication material for the Humanities, Social, and Political Sciences dept. at ETHZ.

EVENTS CHAIR VSETH ETHICS COMMISSION

Sep 2025 – Present

Organising events aimed at STEM students to engage them with the opportunities and impacts of their work and research at ETHZ, including organising a *Technology Ethics Film Festival*.

AI INTERN UNICRI CENTRE FOR AI AND ROBOTICS

Jan 2023 – Aug 2023

Helped to catalogue, discover, and communicate about new AI tools for law enforcement focused on fighting online child abuse through [AI for Safer Children](#) project.

BSC THESIS BAYESIAN NETWORK EXPLAINABILITY

Feb 2022 - Jun 2022

Designed a computational-level theoretical model for encoding a user's level of domain-expertise when generating explanations for Bayesian Network-based decision support systems. ([Gadhia, 2022](#))

Relevant research experience

MSC THESIS *Occupational Gender Bias in LLM Post-Training (Supervised by Dr. E. Ash)* **JUL 2026 (EXPECTED)**



- Developing end-to-end pipeline for synthetic data generation with open-source LLMs focusing on generating narratives for individuals given demographic information.
- Evaluate generation quality disparities relative to gender, education, and profession of individuals including measuring model-uncertainty and diversity.
- Note changes in gender-inference bias through different post-training stages

RESEARCH PAPER *'Right to Work in the age of GenAI' (Supervised by Dr. K. Wodajo)* **OCT 2025**

- Analyse the 'Right to Work' as framed in the Universal Declaration of Human Rights
- Reframe current labour market trends (and potential displacement) as a Human Rights challenge; problematise current conceptualisations of *Work*
- Propose economic policy to engage state and private institutions to address challenge

COURSE PAPER *Managed by AI, Not Replaced (Course: AI Personhood and Social Justice)* **JUN 2025**

- Essay analysing our social conceptions of work as hierarchically high- or low-skill
- Problematising normative narratives of AI-induced labour displacement
- Arguing a more accurate framing of the labour injustice at hand as a removal of autonomy using the case study of out-sourced content moderation to workers in Global South nations

COURSE PROJECT *Social learning through common interests (Course: Design in Education Tech)* **JUN 2025**

- Built a prototype of a tool embedded in Discord (as a test platform) to track students' notes and questions during remote- and hybrid-learning environments
- Did real-time semantic analysis on their notes to group them so that groups consisted of students with either similar questions (to prompt discussions) or with complementary understanding of topics (so they can help one another).

RESEARCH PAPER *OpenHirsh: Detecting Political Narratives with LLMs (Supervised by Dr. E. Ash)* **JUL 2024**

- Using Albert O. Hirschman's taxonomy ("The Rhetoric of Reaction", 1991) for detecting progressive and reactionary political narratives applied to the **Manifesto Corpus** dataset
- Engineered prompts for GPT-4o to classify chunks of text from the dataset to classification task
- Classification result against human annotation: high overall accuracy, skewed by one dominant class, and low F1-score indicating significantly weak recall for minority classes.

BSC THESIS *Bayesian Network Explainability (Supervised by Dr. J. Kwisthout)* **JUN 2022**

- Developed a theoretical cognitive model for Bayesian-network based *decision support systems* to generate "chain of reasoning explanations" for users based on their self-reported level of domain expertise
- Focused on the context of medical diagnostic support systems to be used by doctors (high-level experts), students (medium-level experts), and patients (low-level experts).

PERSONAL PROJECT *PhilGPT RAG* **JUL 2025**

- Built a pipeline to scrape transcripts of the *PhilosophizeThis!* podcast (with creator permission)
- Processed data to build a RAG pipeline for a question-answering system based on the philosophy content using Gemini API for answer generation.